

Are grammatical tones phonetically and diachronically different from lexical tones?
A case study on San Martín Duraznos Mixtec

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In this pilot study, I examine the phonetic realizations of (high) tone in San Martín Duraznos Mixtec (Otomanguean; Mexico) in a corpus of connected speech, asking whether and to what extent morphosyntactic status and diachronic origin condition these realizations. Mixtec languages provide an ideal starting point for such investigations, as tone has a high functional load in lexicon and grammar. High tones, like other tones, are used for marking lexical (ex. 1) and grammatical (ex. 2) distinctions. As can be seen in ex. 2, the imperfective is marked with a high tone on the first mora of the verb stem, a pattern that is found to some extent across all Mixtec varieties studied to date (cf. Costello 2014, Palancar et al. 2016, Swanton & Mendoza Ruiz 2021) and is reconstructable. Building on these observations, I hypothesize that the high tone imperfective marker exhibits systematic phonetic differences compared to high tones used in lexical contrasts.

Based on a corpus of connected speech (Auderset & Hernández-Martínez 2022), I outline an approach to studying phonetic variation in context with an eye to diachrony. The transcriptions are first tokenized and aligned on the phone level, allowing for the extraction and analysis of relevant phonetic measurements (primarily pitch, duration, and intensity). Preliminary results suggest that grammatical high tones tends to be slightly higher on average (Fig. 1) and slightly shorter in duration compared to non-morphemic high tones. Comparing pitch across tonemes is complicated by various language-specific factors such as moraic alignment and glottalization, but also by more general processes like fast-speech reduction and utterance-level prosody. I approach these issues from a diachronic point of view and investigate to what extent sound changes and phonetics can inform each other on the tonal level. While there is a growing body of knowledge on tone change and phonetics in lexical tone (Yang & Xu 2019, Yang et al. 2025), this study is the first to address the issue with grammatical tone.

Examples:

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|----|-------------------------|------------|---------------------|
| 1. | a) <i>ká'àn</i> [kaʔãɬ] | IMPF.speak | ‘is/was speaking’ |
| | b) <i>ká'án</i> [kaʔãɬ] | IMPF.think | ‘is/was thinking’ |
| 2. | a) <i>ká'àn</i> [kaʔãɬ] | IMPF.speak | ‘is/was speaking’ |
| | b) <i>kà'àn</i> [kaʔãɬ] | IRR.speak | ‘will/should speak’ |

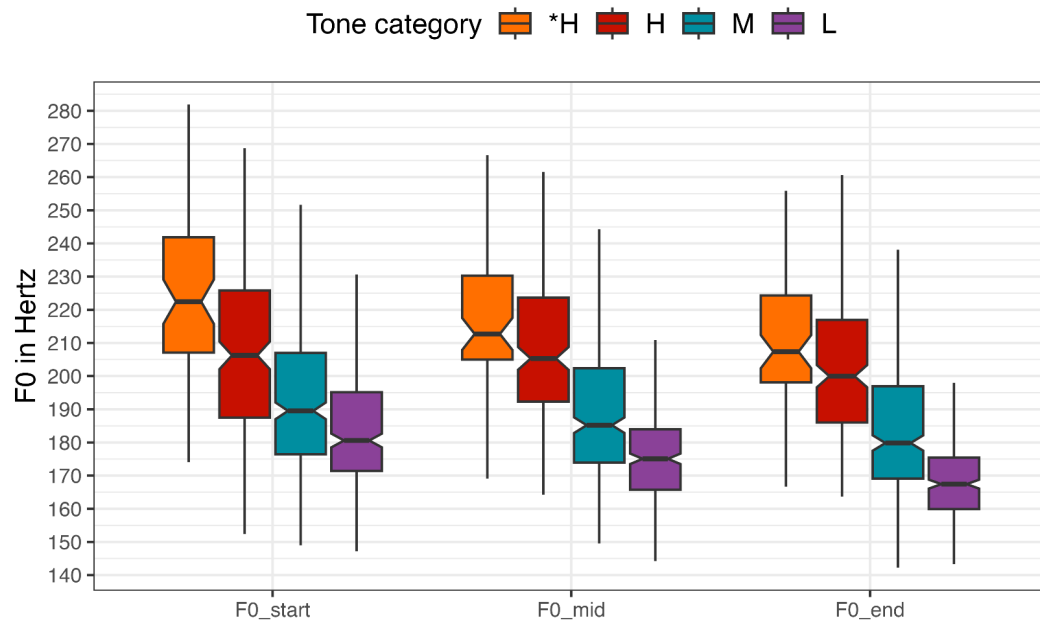


Fig. 1: Pitch measured at three points per short vowel (TBU) grouped per toneme for 1 speaker (1946 tokens). *H = grammatical high tone; H = high tone, M = mid tone, L = low tone (all lexical).

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